Suyeon Lee

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Research Interest

My research focuses on multimodal AI, particularly audio-visual speech processing and source separation. I investigate how cross-modal cues can be leveraged to isolate target signals and improve speech perception in complex acoustic environments, with the goal of building robust multimodal systems for real-world applications.

Education

Korea Advanced Institute of Science and Technology (KAIST) Mar 2025 - Feb 2028 (Expected)
Ph.D. in Electrical Engineering *Advisor: Prof. Joon Son Chung*

Korea Advanced Institute of Science and Technology (KAIST) Mar 2023 - Feb 2025
M.S. in Electrical Engineering *Advisor: Prof. Joon Son Chung*
- Thesis: Multi-modal Speech and Audio Separation System

Korea Advanced Institute of Science and Technology (KAIST) Mar 2018 - Feb 2023
B.S. in Electrical Engineering

Experience

Applied Scientist Intern Nov 2026 - Mar 2027 (Expected)
Amazon Web Services (AWS) *Seattle, WA, United States*

Undergraduate Research Intern Mar 2022 - Feb 2023
Multimodal AI Lab, KAIST *Advisor: Prof. Joon Son Chung*
- Studied generative models, specifically diffusion-based models
- Implemented an active speaker detection baseline

Research Intern Sep 2021 - Feb 2022
AIRS Company, Hyundai Motor Group *Seoul, South Korea*
- Developed wake word detection baseline model for in-car voice assistant

Undergraduate Research Intern Mar 2021 - Aug 2021
SSSC Lab, KAIST *Advisor: Prof. Hoirin Kim*
- Applied fundamental machine learning principles to develop a baseline speech recognition model

Honors and Awards

NIST Speaker Recognition Evaluation (SRE) 2024
1st Place in the Audio Track and 4th Place in the Audio-Visual Track *San Juan, Puerto Rico*

Skills

Programming: PyTorch, Python, C

Academic Activities

Teaching Assistant Fall 2023, Fall 2025
KAIST - EE488: Deep learning for Computer Vision
- Supported students in coding assignments by offering one-on-one debugging and technical guidance

Languages

Korean: Native

English: Proficient

Publications *: Equal Contribution

[9] **SelectSep: Target-Selective Video Queries for Audio Source Separation**
Suyeon Lee, Doyeop Kwak, Joon Son Chung
Preprint. Under review

[8] **I4U's Official and Streamlined Audio Systems for NIST SRE24**
Daniele Colibro, Claudio Vair, Youzhi Tu, Junjie Li, Zilong Huang, Yijia Chen, Kong Aik Lee, Man-Wai Mak, Jagabandhu Mishra, Vishwanath Singh, Xi Xuan, Manasi Chhibber, Oguzhan Kurnaz, Tomi Kinnunen, **Suyeon Lee**, Chaeyoung Jung, Kihyun Nam, Joon Son Chung, Shuai Wang
Odyssey 2026

[7] **LRS-VoxMM: A benchmark for in-the-wild audio-visual speech recognition**
Doyeop Kwak, Jeongsoo Choi, **Suyeon Lee**, Joon Son Chung
Technical Report

[6] **Plug-and-Steer: Decoupling Separation and Selection in Audio-Visual Target Speaker Extraction**
Doyeop Kwak*, **Suyeon Lee***, Joon Son Chung
Interspeech 2026 (Oral)

[5] **Cinematic Audio Source Separation Using Visual Cues**
Kang Zhang*, **Suyeon Lee***, Arda Senocak, Joon Son Chung
IEEE / CVF Conference on Computer Vision and Pattern Recognition 2026

[4] **Model-Guided Dual-Role Alignment for High-Fidelity Open-Domain Video-to-Audio Generation**
Kang Zhang*, Trung X. Pham*, **Suyeon Lee**, Axi Niu, Arda Senocak, Joon Son Chung
Conference on Neural Information Processing Systems 2025

[3] **FlowAVSE: Efficient Audio-Visual Speech Enhancement with Conditional Flow Matching**
Chaeyoung Jung, **Suyeon Lee**, Ji-Hoon Kim, Joon Son Chung
Interspeech 2024

[2] **Seeing Through the Conversation: Audio-Visual Speech Separation based on Diffusion Model**
Suyeon Lee*, Chaeyoung Jung*, Youngjoon Jang, Jaehun Kim, Joon Son Chung
IEEE International Conference on Acoustics, Speech, and Signal Processing 2024 (Oral)

[1] **TalkNCE: Improving Active Speaker Detection with Talk-aware Contrastive Learning**
Chaeyoung Jung*, **Suyeon Lee***, Kihyun Nam, Kyeongha Rho, You Jin Kim, Youngjoon Jang, Joon Son Chung
IEEE International Conference on Acoustics, Speech, and Signal Processing 2024 (Oral)